Voltage Divider and Current Divider in LTspice

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1 Introduction

In this report, I simulate a voltage divider in LTspice and compare the results with hand calculations. I then connect a load to the output to see how it changes the output voltage. Finally, I study a current divider and verify it in LTspice.

2 Voltage Divider Rule

The output voltage is

$$V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$$

With $V_{in} = 9\,V$, $R_1 = 10\,k\Omega$ and $R_2 = 4.7\,k\Omega$:

$$V_{out} = 9 \cdot \frac{4.7}{10 + 4.7} = 2.878\,V$$

2.1 LTspice Simulation

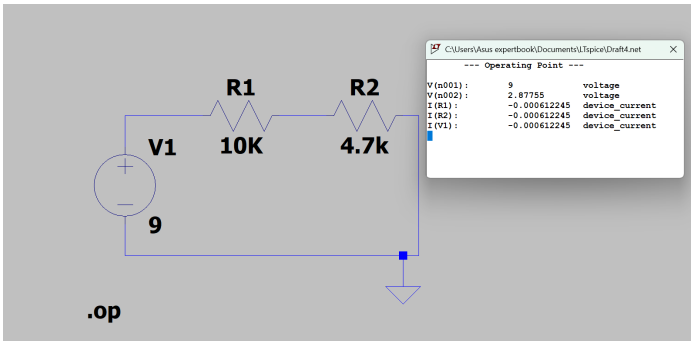


Figure 1: Voltage divider without load

3 Voltage Divider with a Load

A load $R_3 = 1\,k\Omega$ is connected in parallel with R_2:

$$R_{23} = R_2 \parallel R_3 = \frac{R_2 \cdot R_3}{R_2 + R_3} = \frac{4.7 \times 1}{4.7 + 1} \approx 0.825\,k\Omega$$

$$V_{out} = V_{in} \cdot \frac{R_{23}}{R_1 + R_{23}} = 9 \times \frac{0.825}{10 + 0.825} \approx 0.686\,V$$

3.1 LTspice Simulation

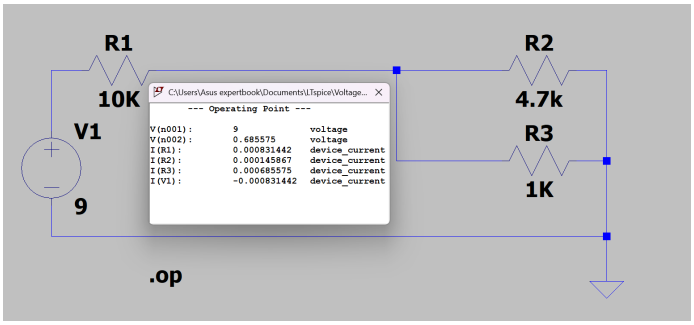


Figure 2: Voltage divider with a $1\,k\Omega$ load

4 Notes :

we observed that the voltage drops because of the low-value load. To solve this problem, the load must be much larger than R_2, so that the output voltage stays close to its theoretical value.

$$let : R_3 = 470k\Omega$$

so :

$$R_{23} = \frac{R_2 \cdot R_3}{R_2 + R_3} \approx R_2$$

$$V_{out} = 9 \times \frac{4.7}{10 + 4.7} \approx 2.8775V$$

5 Current Divider law :

The equivalent Resistance :

$$R_{eq} = R_1 + \frac{R_2 \cdot R_3}{R_2 + R_3} = 1 + \frac{2 \times 3}{2 + 3} = 2.2k\Omega$$

The total current is :

$$I_t = \frac{V_{in}}{R_{eq}} = \frac{12}{2.2} \approx 5.45mA$$

The current in R_2 and R_3 :

$$I_2 = I_t \cdot \frac{R_3}{R_2 + R_3} = 5.45 \times \frac{3}{2+3} \approx 3.27mA$$

$$I_3 = I_t \cdot \frac{R_2}{R_2 + R_3} = 5.45 \times \frac{2}{3+2} \approx 2.18mA$$

Check using Kirchhoff's Current Law (KCL):

$$I_2 + I_3 = 3.27\,mA + 2.18\,mA = 5.45\,mA = I_t$$

6 On LTspice :

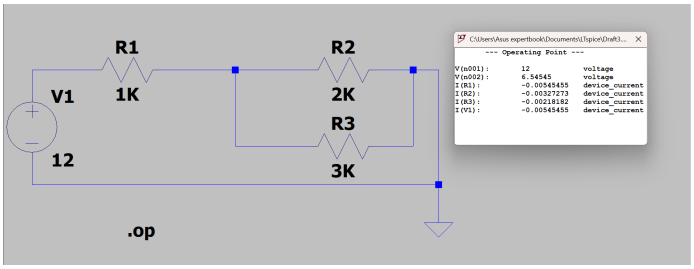


Figure 3: Current Divider